CS G516: Advanced Database Systems

Database Systems Studio

Spatio-Temporal Mobility Database

Designing and Engineering a Database for Moving Entities

---

1. Studio Overview

The **Database Systems Studio** is the project-based component of **CS G516: Advanced Database Systems**. Throughout the semester, teams will design, implement, query, and optimise a database for a **mobility system whose entities move through space and change over time**. The project focuses on **database engineering**, including:

- data and workload modelling;
- spatial and temporal data;
- advanced querying;
- indexing and physical design;
- query optimisation;
- performance evaluation.

Building a frontend or full-stack application is **not required**.

2. Project Theme

Spatio-Temporal Mobility Database

> **Design a database system for storing, understanding, and querying moving entities in space and time.**

Each team will choose one mobility scenario.

A. Campus shuttle / e-vehicle

Vehicles, trips, routes, stops, GPS positions, speed, occupancy and delays.

B. Campus bicycle / e-bike sharing

Bikes, users, stations, trips, GPS trajectories, timestamps and battery information.

C. Campus food-delivery / logistics

Delivery vehicles, orders, origins, destinations, trajectories and delivery events.

D. Campus waste collection

Collection vehicles, bins, locations, collection events, routes and waste quantities.

E. Campus security / patrol

Patrol vehicles/teams, trajectories, incidents, locations and response events.

3. Project Objective

The project should progress beyond simply storing and retrieving records.

Your database should enable questions such as:

Spatial

Which vehicles are within 500 m of a particular location?

Temporal

Which vehicles were active between 8:00 and 9:00 AM?

Spatio-temporal

Which vehicles entered Zone A between 8:00 and 8:30 AM?

Movement / trajectory

Which trips followed similar movement patterns?

Analytical

How does mobility demand vary across regions and time?

You will investigate how **data representation, query processing, indexing, and physical design** affect the ability to answer such questions efficiently.

4. Common Project Requirements

Regardless of the chosen scenario, every team must:

1. Model moving entities

Represent entities such as vehicles, bikes, robots, or patrol units and their movement history.

2. Represent space

Represent appropriate spatial objects such as:

- points;
- routes;
- regions;
- stations;
- buildings;
- trajectories.

3. Represent time

Support:

- timestamps;
- time intervals;
- durations;
- historical information.

4. Implement the database

Develop a working database containing the required data and constraints.

5. Implement an advanced query workload

The final system must include:

- basic operational queries;
- spatial queries;
- temporal queries;
- spatio-temporal queries;
- analytical queries.

6. Investigate performance

Evaluate the effect of appropriate:

- indexes;
- query formulations;
- physical design choices;
- database optimisation techniques.

7. Customisation

The project has a common technical core, but teams have scope for customisation.

You may customise:

- the mobility scenario;
- domain-specific entities and events;
- additional data;
- additional queries;
- the advanced investigation;
- the final research question.

Your customisation should add **database complexity or insight**, not merely application features.

8. Project Development Milestones

The project will be developed incrementally through the following milestones.

Milestone	Focus	Key deliverable
M1	**Problem & Workload Analysis**	Mobility-system definition, entities, events, spatial/temporal data and representative workloads
M2	**Data & Database Design**	Conceptual model, logical schema, spatial/temporal representation and design justification
M3	**Database Implementation**	Working database, data loading, constraints and baseline queries
M4	**Advanced Querying**	Spatial, temporal, spatio-temporal and analytical queries
M5	**Indexing & Optimisation**	Index design, query plans and performance experiments
M6	**Custom Investigation**	Team-defined question, database-based analysis and findings
M7	**Final Evaluation**	Integrated system, performance evaluation, technical report and demonstration

These milestones are **development checkpoints**. Teams are expected to progressively refine their work throughout the semester.

9. Formal Project Assessment

The Database Systems Studio contributes **25%** to the course grade.

Assessment 1 — Design Review: 10%

Conducted before the Mid-Semester Examination.

The Design Review will assess:

- problem and workload analysis;
- requirements;
- conceptual/data model;
- database schema;
- spatial and temporal representation;
- initial database design;
- justification of design decisions.

The team should demonstrate that it has a **well-defined problem, workload, and technically defensible database design**.

Assessment 2 — Final Demonstration & Technical Report: 15%

Conducted at the end of the semester.

The Final Evaluation will assess:

- working database system;
- advanced query workload;
- spatial, temporal and spatio-temporal functionality;
- indexing and optimisation;
- performance experiments;
- custom investigation;
- technical understanding and justification;
- final technical report and demonstration.

A working database alone is not sufficient. You must be able to explain **why your design works, where it performs well or poorly, and why.**

10. What Is Not Required

You do **not** need to build:

- a web application;
- a mobile application;
- a graphical user interface;
- a REST API;
- an authentication system.

The primary deliverables are the **database, queries, experiments, analysis, and technical justification**.

11. Central Design Question

Throughout the project, keep asking:

What database design is appropriate for this data and this workload—and why?

The objective of the Studio is not simply to build a database.

It is to learn how to **design, engineer, measure, and defend a database system for a non-trivial workload**.

Technology & Tools

The Studio will use **PostgreSQL + PostGIS** as the primary database platform.

Students may use additional tools as appropriate for their project.

Core technology

- **PostgreSQL** — relational database management system
- **PostGIS** — spatial data and spatial query support
- **SQL** — database definition, querying and analysis

Supporting tools

Students may independently explore tools for:

- data generation and loading;
- scripting and automation;
- performance measurement;
- data analysis and visualisation.

For example, **Python** may be used where appropriate.

> **The course will teach the underlying database concepts and techniques; students are expected to learn the relevant tools and implementation details independently as required by their project.**

Important

Tool choice is **not the objective of the Studio**. Teams will be evaluated primarily on their:

database design → query workload → implementation → optimisation → experimental evaluation → technical reasoning